Day 16 5/11/26

88. Merge Sorted Array

nums1 = [1,2,3,0,0,0], m=3, nums2 = [2,5,6], n=3
output [1,2,2,3,5,6]

Approach:- Three Pointers from the End

The important idea is to start from the end, not from the begining.

nums1 already has extra empty spaces (0s) at the end. If we start from the begining, we may overwrite useful values.

Code :-

```
class Solution{
  public void merge(int[] nums1, int m, int[] nums2, int n)
  {
    int i=m-1, j=n-1, k=m+n-1;
    while (i>=0 && j>=0){
      if(nums1[i] > nums2[j]){
        nums1[k] = nums1[i];
        i--;
      }else{
        nums1[k] = nums2[j];
        j--;
      }
      k--;
    }
    while(j>=0){
      nums1[k] = nums2[j];
      j--, k--;
    }
  }
}
```

Time Complexity:
O(m+n)
Space Complexity:
O(1)

Description | Accepted | Editorial | Solutions | Subm

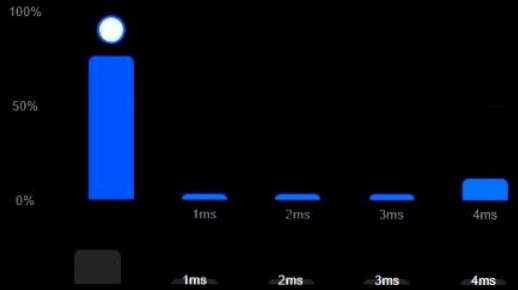
All Submissions

Runtime

0 ms | Beats 100.00%

Memory

43.17 MB | Beats 92.75%



Code | Java

Code

Java | Auto

```java
class Solution {
    public void merge(int[] nums1, int m, int[] nums2, int n) {

        int p1 = m - 1, p2 = n - 1, i = m + n -1;
        while(p2>=0){
            if(p1 >= 0 && nums1[p1] > nums2[p2]){
                nums1[i] = nums1[p1];
                p1--;
            }else{
                nums1[i] = nums2[p2];
                p2--;
            }
            i--;
        }
```

Saved

Ln 4, Col 50

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums1 =